

NORTH AMERICAN ELECTRIC RELIABILITY CORPORATION

# NERC RC CERTIFICATION EXAM STUDY GUIDE

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Reliability Coordinator (RC)  
Comprehensive Exam Preparation

Covering All 10 Study Chapters:  
BAL · EOP · INT · IRO · TOP · VAR · PRC · FAC · COM · MOD

**Ricardo Barbosa**  
NERC-Certified EDF Operator

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# Chapter

## RC Certification Overview

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### 1.1 What is the Reliability Coordinator?

The Reliability Coordinator (RC) is the entity with the widest area view of, and the highest level of authority over, the Bulk Electric System (BES). The RC monitors the real-time operating state of its RC Area and coordinates corrective actions to maintain reliability—even if those actions override decisions made by Transmission Operators, Balancing Authorities, and Generator Operators.

#### Key Concept

The RC is the highest operating authority in the NERC reliability hierarchy. When the RC issues an Operating Instruction, subordinate entities must comply—or immediately inform the RC of their inability to do so. This authority derives from **IR0-001-4**, the foundational standard for RC operations.

### 1.2 NERC Certification Programs

NERC administers five system operator certifications. The RC certification is the most comprehensive, covering wide-area reliability oversight, emergency coordination, and all major reliability standard domains.

| Certification | Full Name                 | Scope                                              |
|---------------|---------------------------|----------------------------------------------------|
| RC            | Reliability Coordinator   | Wide-area reliability; highest operating authority |
| BA            | Balancing Authority       | Real power balancing, ACE, reserves                |
| TOP           | Transmission Operator     | Real-time transmission system operation            |
| GOP           | Generator Operator        | Real-time generation unit operation                |
| BI            | Balancing and Interchange | Combined BA + interchange scheduling               |

### WARNING

**ISO is NOT a NERC certification.** ISO (Independent System Operator) is a type of market entity—CAISO, MISO, PJM. It is not a NERC operator credential. This is a common distractor in exam questions. Do not confuse the two.

## 1.3 Exam Format

| Parameter                      | Value       |
|--------------------------------|-------------|
| Questions (scored)             | 100–120     |
| Questions (unscored pilot)     | ~20         |
| Duration                       | 3–4 hours   |
| Administrator                  | Pearson VUE |
| Scenario-based questions       | ~65%        |
| Direct knowledge questions     | ~25%        |
| Calculation questions          | ~10%        |
| Passing score                  | ~70%        |
| Renewal cycle                  | 3 years     |
| Continuing education (renewal) | 36 hours    |

## 1.4 The Four Exam Domains

#### 1.4. THE FOUR EXAM DOMAINS

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| Domain | Name                                 | Content                                              |
|--------|--------------------------------------|------------------------------------------------------|
| 1      | Real-Time Operations                 | ACE, AGC, frequency response, reserves, CPS1/CPS2    |
| 2      | Transmission Operations              | SOLs, IROLs, voltage/reactive, TLR procedures        |
| 3      | Emergency Operations                 | Emergency response, restoration, blackstart, UFLS    |
| 4      | Situational Awareness & Coordination | OPA, wide-area monitoring, RC authority, interchange |



## Chapter 2

# Resource & Demand Balancing

### 2.1 The ACE Formula

Area Control Error (ACE) is the primary metric for real-time balancing. ACE must be maintained near zero to meet both interchange obligations and frequency support duties.

#### Formula: Area Control Error (ACE)

$$ACE = (NIA - NIS) - 10B(F_A - F_S) - IME$$

|       |                                                         |
|-------|---------------------------------------------------------|
| NIA   | Net Interchange Actual (measured MW, positive = export) |
| NIS   | Net Interchange Scheduled (contracted MW)               |
| B     | Frequency Bias Setting (MW/0.1 Hz; always negative)     |
| $F_A$ | Actual Interconnection Frequency (Hz)                   |
| $F_S$ | Scheduled Frequency (normally 60.00 Hz)                 |
| IME   | Inadvertent Meter Error correction                      |
| 10    | Unit conversion: MW/0.1Hz $\rightarrow$ MW/Hz           |

#### Exam Tip

The most common calculation error: forgetting the factor of 10 in the bias term. B is in MW/0.1 Hz. The formula uses it in MW/Hz. Missing the factor of 10 gives answers 10 $\times$  off, which eliminates correct options and highlights wrong ones.

### 2.2 Reserve Requirements

## Operating Reserve Components

Operating Reserve = Regulation Reserve + Contingency Reserve

- **Regulation Reserve:** Used by AGC in real time to control ACE. Fast-responding, continuously adjustable units under automatic control.
- **Contingency Reserve:** Held for the Most Severe Single Contingency (MSSC). Must be  $\geq$  MSSC at all times. Subdivided into:
  - *Spinning Reserve:* Online, synchronized; deployable in 10 minutes.
  - *Non-Spinning Reserve:* Offline; also deployable in 10 minutes.

### BAL-002-3: Disturbance Control Standard

| Requirement                     | Value                                  |
|---------------------------------|----------------------------------------|
| Contingency Reserve minimum     | $\geq$ MSSC (MW)                       |
| ACE recovery after contingency  | Within 15 minutes (to within DRL)      |
| Reserve restoration             | Within 90 minutes of contingency start |
| Spinning Reserve deployment     | Within 10 minutes                      |
| Non-Spinning Reserve deployment | Within 10 minutes                      |

## 2.3 Time Error Corrections

| Condition       | Clocks Running            | BAs Set $F_S$ to |
|-----------------|---------------------------|------------------|
| Slow Time Error | Slow (freq avg $< 60$ Hz) | 60.02 Hz         |
| Fast Time Error | Fast (freq avg $> 60$ Hz) | 59.98 Hz         |

## Chapter

# Transmission Operations

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### 3.1 SOL vs IROL

**SOL (System Operating Limit):** The value of a system parameter (MW, voltage, frequency) that, if exceeded, violates applicable facility ratings, voltage limits, or system stability limits. Thousands of SOLs exist on large systems.

**IROL (Interconnection Reliability Operating Limit):** A subset of SOLs whose violation could adversely affect the reliability of two or more RC Areas in the Interconnection. Each IROL has a violation time  $T_v \leq 30$  minutes.

#### **Critical — Immediate Action Required**

When an IROL has been violated beyond its  $T_v$  (maximum 30 minutes), the RC **MUST** immediately take or direct emergency actions. Acceptable actions include: directing load shedding, curtailing interchange, reconfiguring transmission, or directing emergency generation dispatch. No further grace period applies.

### 3.2 TLR Levels (Eastern Interconnection)

| Level | Action               | What is Curtailed                         |
|-------|----------------------|-------------------------------------------|
| 1     | Notify               | Notification only; no curtailment         |
| 2     | Reconfigure          | Redirect/reallocate non-firm transactions |
| 3a    | Curtail Redirects    | Curtail non-firm redirected transactions  |
| 3b    | Curtail Non-Firm PTP | Curtail non-firm Point-to-Point service   |
| 4     | Curtail Firm PTP     | Curtail firm Point-to-Point service       |
| 5a    | Curtail All Non-Firm | All remaining non-firm service            |
| 5b    | Curtail Firm Network | Network Integration Transmission Service  |

### 3.3 Power Flow Fundamentals

#### Formula: Real and Reactive Power Flow

$$P \approx \frac{V_1 V_2}{X} \sin(\delta_1 - \delta_2) \quad (\text{MW flow driven by angle difference})$$

$$Q \approx \frac{V_1 V_2}{X} \cos(\delta) - \frac{V_2^2}{X} \quad (\text{MVAR flow driven by voltage magnitude difference})$$

## Chapter

# Emergency Preparedness

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### 4.1 IRO-008-2: Operational Planning Analysis (OPA)

The RC must perform a next-day OPA assessing whether planned operations will violate SOLs or IROLs, identify anticipated capacity deficiencies, and coordinate solutions with applicable TOPs and BAs before the operating day begins.

### 4.2 EOP-011-2: Emergency Operating Plans

Each Transmission Operator must maintain documented Operating Plans for:

- Capacity deficiency emergencies
- Voltage emergencies
- Transmission overload conditions
- Severe weather and geomagnetic disturbance events

### 4.3 EOP-005-3: Blackstart Resources

#### Exam Tip

EOP-005-3 requires Generator Operators with blackstart resources to provide training to operating personnel every **2 calendar years**. This is a critical exam number— not annually, not every 3 years. The exam tests this specifically.



# Chapter 5

## Emergency Response & Restoration

### 5.1 System Restoration: Blackstart Basics

#### Cranking Path

A Cranking Path is a pre-determined switching sequence establishing an energized transmission path from a blackstart resource to one or more generating units to be restarted. Must be documented in advance under EOP-005-3. Operators follow the Cranking Path during restoration—improvising energized switching creates new faults.

#### Cold Load Pickup Risk

Cold load (load that has been de-energized) draws 2–5× normal inrush current when re-energized, as motors, thermostats, and heaters all demand maximum power simultaneously. On weak restoration islands, this frequency collapse can re-trip the restoration generators. **Pick up load in small, carefully controlled increments.**

### 5.2 UFLS and Frequency Thresholds

| Frequency | System State                                     |
|-----------|--------------------------------------------------|
| 60.00 Hz  | Nominal — AGC managing ACE                       |
| 59.95 Hz  | Below normal — governor response activating      |
| ~59.3 Hz  | UFLS begins automatically (PRC-006 settings)     |
| 59.0 Hz   | Severe — generator trips, system separation risk |

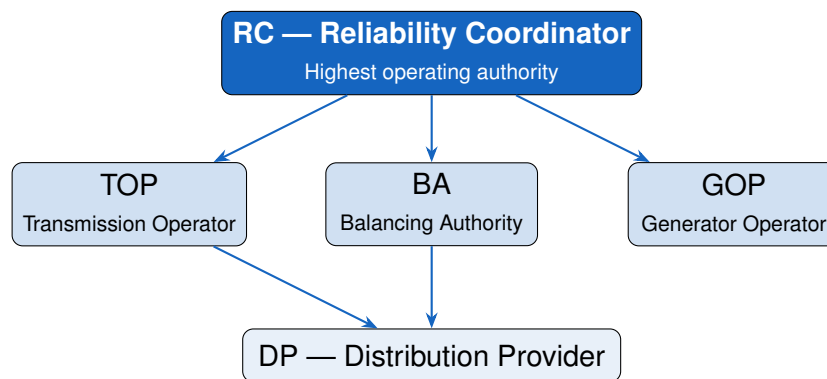


## Chapter 6

# IRO — RC Authority & Wide-Area Reliability

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## 6.1 The Operating Hierarchy



## 6.2 IRO-001-4: RC Authority

Under IRO-001-4, the RC:

- Shall take corrective actions to maintain the reliability of its RC Area
- Shall issue Operating Instructions to subordinate entities (TOPs, BAs, GOPs, DPs)
- Shall coordinate with adjacent RCs when wide-area conditions require it

Subordinate entities must:

- Comply with RC Operating Instructions

- **Immediately inform the RC if they cannot comply**—silence is a violation
- Coordinate planned outages with the RC in advance

## Chapter 7

# Interchange Scheduling

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### 7.1 E-Tag Required Elements

| Element            | Description                                 |
|--------------------|---------------------------------------------|
| MW Amount          | The amount of energy to be transferred      |
| Start/End Time     | Operating period for the interchange        |
| Ramp Time and Rate | Beginning/ending ramp times and MW/min rate |
| Type of Service    | Firm or Non-Firm for delivery and receipt   |
| Source and Sink    | Generating resource and receiving load area |
| Transmission Path  | The path over which the energy flows        |

### 7.2 INT-006-5: Reliability Adjustment

#### Exam Tip

When the RC directs modification of Confirmed or Implemented Interchange for reliability reasons, a Reliability Adjustment Arranged Interchange schedule must be submitted within **60 minutes** of the start of the modification. This is a frequently tested timing requirement.



## Chapter

# Power Systems Fundamentals

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### 8.1 Governor Droop Response

#### Formula: Governor Droop Calculation

$$\Delta \text{MW} = \frac{\text{Frequency Deviation (\%)}}{\text{Droop (\%)}} \times \text{Unit Rating (MW)}$$

**Example:** 5% droop, 100 MW unit, frequency drops from 60.00 to 59.70 Hz

$$\text{Deviation} = \frac{0.30}{60} \times 100\% = 0.5\% \quad \Rightarrow \quad \Delta \text{MW} = \frac{0.5\%}{5\%} \times 100 \text{ MW} = 10 \text{ MW}$$

### 8.2 Three Levels of Frequency Control

**Primary (seconds):** Autonomous governor response, proportional to frequency deviation. Arrests frequency decline but does not restore 60 Hz.

**Secondary (minutes):** AGC adjusts unit setpoints to eliminate ACE and restore frequency to scheduled value (60 Hz normally).

**Tertiary (10–30 min):** Reserve restoration—start additional units, arrange emergency assistance, replenish contingency reserves.



# Chapter 9

## Voltage & Reactive Power

### 9.1 MVAR Sources and Sinks

| Device/Condition       | Effect on MVAR                              |
|------------------------|---------------------------------------------|
| Overexcited generator  | Produces MVAR (raises voltage)              |
| Underexcited generator | Absorbs MVAR (lowers voltage)               |
| Capacitor bank         | Produces MVAR (raises voltage)              |
| Shunt reactor          | Absorbs MVAR (lowers voltage)               |
| Heavily loaded line    | Consumes MVAR (drops receiving-end voltage) |
| SVC / STATCOM          | Adjustable MVAR injection or absorption     |

### 9.2 VAR Standards Summary

**VAR-001-6:** Transmission Operator establishes voltage schedules or reactive power output schedules for generators at their interconnection points.

**VAR-002-4:** Generator Operators must maintain terminal voltage or reactive output within schedules set by the TOP. If unable to comply, must immediately notify the TOP.



# Chapter 10

## Practice & Exam Strategy

### 10.1 Critical Numbers Reference

| Value             | Meaning                                         | Standard          |
|-------------------|-------------------------------------------------|-------------------|
| 15 minutes        | DRL recovery after contingency                  | BAL-002-3         |
| 90 minutes        | Reserve restoration after contingency           | BAL-002-3         |
| 10 minutes        | Spinning/Non-Spinning Reserve deployment        | BAL-002-3         |
| $\leq 30$ minutes | Maximum IROL violation time (Tv)                | NERC Glossary     |
| 60 minutes        | Reliability Adjustment e-tag submission         | INT-006-5         |
| 2 years           | Blackstart personnel training cycle             | EOP-005-3         |
| 3 years           | RC certification renewal cycle                  | NERC Cert Program |
| 36 hours          | CE hours required for 3-year renewal            | NERC Cert Program |
| 60.02 Hz          | Scheduled frequency, slow time error correction | BAL-004           |
| 59.98 Hz          | Scheduled frequency, fast time error correction | BAL-004           |
| $\geq 100\%$      | CPS1 minimum performance standard               | BAL-001-2         |
| $\sim 59.3$ Hz    | Typical UFLS first step threshold               | PRC-006           |

### 10.2 Scenario Question Framework

For every scenario-based question, apply this four-step framework:

1. **Who has authority?** Identify the responsible entity: RC, TOP, BA, GOP. The RC has widest authority but directs, not operates, equipment.
2. **What standard applies?** Map the scenario: frequency  $\rightarrow$  BAL; voltage  $\rightarrow$  VAR/EOP; overload  $\rightarrow$  TOP/IRO; blackstart  $\rightarrow$  EOP-005.

3. **What is the required timeline?** Reference the table above. Timelines are frequently tested and often distinguish correct from incorrect answers.
4. **What is the correct sequence?** Assess → coordinate → direct action. The RC rarely acts unilaterally—it assesses, then directs the appropriate entity.

### WARNING

#### Common Mistakes on the RC Exam:

- Selecting "shed load immediately" before assessing — RC assesses first
- Confusing ISO (market entity) with a NERC certification
- Mixing up fast/slow time error scheduled frequency values
- Forgetting the  $10\times$  factor in the ACE bias term
- Applying TLR procedures to the Western Interconnection (TLR is Eastern only)

## Quick Reference: Key Standards

| Standard       | Topic                                                              |
|----------------|--------------------------------------------------------------------|
| BAL-001-2      | CPS1/CPS2 real power balancing performance                         |
| BAL-002-3      | Disturbance Control Standard — MSSC reserves, DRL, 15-min recovery |
| BAL-003-2      | Frequency response and bias settings                               |
| BAL-005        | Balancing Authority control — AGC requirements                     |
| BAL-006        | Inadvertent interchange accounting                                 |
| COM-001        | Communications — primary and alternate paths                       |
| EOP-005-3      | System restoration from blackstart — 2-year training               |
| EOP-006-2      | System restoration coordination — RC role                          |
| EOP-011-2      | Emergency operations preparedness — TOPs maintain plans            |
| FAC-001        | Facility connection requirements                                   |
| FAC-002        | Facility connections and planning analysis                         |
| INT-006-5      | Interchange evaluation and reliability adjustment                  |
| INT-009-3      | RC interchange evaluation                                          |
| IRO-001-4      | RC authority — highest operating authority                         |
| IRO-006-EAST-2 | TLR procedures — Eastern Interconnection                           |
| IRO-008-2      | Operational Planning Analysis (OPA) — next-day                     |
| IRO-017-1      | Wide-area situational awareness tools                              |
| MOD-001        | Generator real and reactive capability data                        |
| PRC-006        | Under-Frequency Load Shedding (UFLS)                               |
| TOP-001-5      | Transmission operations — SOL/IROL compliance                      |
| VAR-001-6      | Voltage and reactive control — TOP sets schedules                  |
| VAR-002-4      | Generator reactive capability — GOPs follow TOP schedules          |